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Study of the One-Dimensional Transverse Ising Model in Open Quantum Systems

Kevin Josué Martínez Herrera
School of Physics, University of Costa Rica
Computational Statistical Mechanics
Instructor: Marlon Brenes

1. Research Question

How can Monte Carlo be applied to study an open quantum system?

The study of open quantum systems constitutes one of the most relevant problems in modern physics, as it allows understanding how the interaction between a system and its environment leads to decoherence and dissipation phenomena that are essential for a realistic description of quantum matter. However, the inherent complexity of the Hilbert space grows exponentially with the number of particles [1], making deterministic methods unfeasible. In this context, Monte Carlo methods offer a powerful tool by allowing the stochastic sampling of representative configurations, drastically reducing computational cost without losing statistical rigor. Its application to the Ising model—one of the paradigmatic systems for studying phase transitions and magnetism—allows exploring the thermodynamics and equilibrium dynamics efficiently, even in the presence of dissipative effects described by the Lindblad master equation. Thus, Monte Carlo becomes not only a numerical method of choice but also a fundamental way to link computational statistical mechanics with quantum physics out of equilibrium. Going into more detail, the methodology performed in the project corresponds to that of a "Path Integral Monte Carlo" (PIMC) process [2], which generally constitutes the finite-temperature numerical method with which systems with a small or abundant number of particles can be studied. It has a wide variety of applications, and in particular it stands out in the study of physical models such as the Ising model.

2. Objectives

2.1. General Objective

Simulate an open quantum system under the Lindblad master equation.

2.2. Specific Objectives

1. Formally describe the dynamics of an open quantum system under the Lindblad master equation
2. Analyze the one-dimensional dissipative Ising model. To achieve this, the main classical and transverse models in closed systems are studied in turn
3. Implement Metropolis-Hastings to compute observables
4. Study the convergence and thermal equilibrium of observables under high and low temperature regimes

3. Theoretical Framework

Markovian Open Quantum Systems

In reality, it is rare for a system to be completely isolated from its environment. Therefore, open quantum systems are defined as those that interact with the environment, such that relaxation and decoherence processes occur that cannot be described by the standard Schrödinger equation applied in closed systems [1]. In the regime of weak coupling with a Markovian environment, i.e., without memory, the dynamics of the reduced density operator can be defined as governed by the Lindblad

master equation. However, before the Lindblad master equation, it is of utmost importance to understand the formal definition of the density matrix in this context.

First, the problem must be stated. A quantum system A is considered, which interacts with an environment B. It will be assumed that A + B is a closed system, such that its evolution is unitary and governed by a Hamiltonian $H = H_A + H_B + V$, with V being the interaction. Then, the total state at t_0 is defined as $\rho(t_0)$; for a time $t_1 \geq t_0$, the total state is $\rho(t_1) = U(t_1, t_0)\rho(t_0)U^\dagger(t_1, t_0)$. Now, as observers, we only have access to system A, whose reduced state is obtained by a partial trace over environment B:

$$\rho_A(t_1) = \text{Tr}_B[\rho(t_1)] = \text{Tr}_B[U(t_1, t_0)\rho(t_0)U^\dagger(t_1, t_0)] \quad (1)$$

3.1. The Dynamical Map and the UDM

A transformation, or a map, will be established such that the evolution of system A can be directly described as follows [3]:

$$\rho_A(t_1) = \varepsilon(t_1, t_0)[\rho_A(t_0)] \quad (2)$$

However, when considering the most general case, the described map is problematic due to the possibility of correlations in the initial state of the form:

$$\rho(t_0) = \rho_A(t_0) \otimes \rho_B(t_0) + \rho_{\text{corr}}(t_0) \quad (3)$$

Now, in order to model the process as Markovian, the physical assumption is made that there is no correlation between system A and its environment B, with which the universal dynamical map (UDM) is defined [4]. Imposing this condition, the map $\varepsilon(t_1, t_0)$ becomes linear and independent of the state on which it acts. Furthermore, it must preserve the properties of the system on which it acts: hermiticity, unit trace, and positive semi-definiteness [3]. Likewise, the study of the evolution of a subsystem within the environment is also relevant, where the density matrix ρ itself evolves; in that case, positivity must be preserved for the entire matrix and not just the subsystem, which is called Complete Positivity (CP).

3.1.1. Choi-Kraus Theorem and UDM Identities

Let $\mathcal{B}(\mathcal{H})$ be the set of bounded linear operators, a linear map ε on $\mathcal{B}(\mathcal{H})$ is completely positive only if:

$$\varepsilon(\rho) = \sum_{\mu} K_{\mu} \rho K_{\mu}^{\dagger}, \forall \rho \in \mathcal{B}(\mathcal{H}) \quad (4)$$

Where K_{μ} are the Kraus operators, which satisfy the following condition:

$$\sum_{\mu} K_{\mu} K_{\mu}^{\dagger} = \mathbb{1} \quad (5)$$

Now, returning to the case of interest, which is the study of system A under the transformation of a UDM, its most general form is [5]:

$$\rho_A(t_1) = \varepsilon(t_1, t_0)[\rho_A(t_0)] = \sum_{\mu} K_{\mu}(t_1, t_0) \rho_A(t_0) K_{\mu}^{\dagger}(t_1, t_0) \quad (6)$$

Where the map ε is completely positive and trace-preserving.

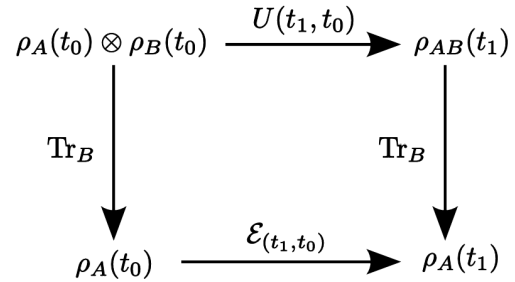


Figure 1: Schematic illustration of the Universal Dynamical Map (UDM), its relation to the initial uncorrelated state and the unitary transformation

3.1.2. Markovian Divisibility

Note that the previous UDM was described only starting at t_0 with a factorized state [4]. However, for the total state $\rho(t_1)$ it is no longer factorized, so there could be correlation between A and B; in that case, $\varepsilon(t_2, t_1)$ is not a UDM. However, similar to what was done for the analysis of the initial state, this map will also be established as Markovian, so it is assumed that no correlations exist for this state. In summary, both for the transition from the initial state to a later one or from a state different from the initial to a later one, it is assumed that no correlations exist. By establishing these conditions, the short-memory nature idiosyncratic to Markov processes is satisfied.

Recall from classical Markov chain processes that a transition matrix Q is presented and the evolving object is described with a probability vector. In this model, the evolution rule with short memory follows the form $\vec{P}(t+1) = Q\vec{P}(t)$, or written in terms of matrix multiplication: $P(t+s) = P(t)P(s)$; formally, this is called the Chapman-Kolmogorov equation.

Now, in the case of a quantum Markov process, the divisibility condition is satisfied, which is analogous to the Chapman-Kolmogorov equation [4]. It states that a quantum process is Markovian if the family of dynamical maps obeys the composition law:

$$\varepsilon(t_2, t_0) = \varepsilon(t_2, t_1)\varepsilon(t_1, t_0), \forall t_2 \geq t_1 \geq t_0 \quad (7)$$

3.2. Generator Matrix

Building on the Chapman-Kolmogorov and continuous Markov setting, this is also described by the generator matrix G [6]. Consider a chain $X(t)$ with an initial condition $X(0) = i$. For an infinitesimal jump $\delta < 0$, where $T_1 \approx \text{Exponential}(\lambda_i)$, the associated probability is:

$$P(T_1 < \delta) = 1 - e^{-\lambda_i} \approx 1 - (1 - \lambda_i \delta) = \lambda_i \quad (8)$$

Then, this λ_i is defined as the transition rate out of state i

$$\lambda_i = \lim_{\delta \rightarrow 0^+} \left[\frac{P(X(\delta) \neq i | X(0) = i)}{\delta} \right] \quad (9)$$

From these definitions, and denoting the transition from $i \rightarrow j$ with probability p_{ij} , the generator matrix can be defined with elements:

$$g_{ij} = \begin{cases} \lambda_i p_{ij}, & i \neq j \\ -\lambda_i, & i = j \end{cases} \quad (10)$$

Now, the stationary state follows the same form of invariance under a linear transformation observed in discrete systems.

$$\pi = \pi P(t) \quad (11)$$

Now, the behavior of the generator matrix can be described by the forward and backward Kolmogorov equations. The backward equation describes the probability of transitioning from one state to another at future times, given the present time [7]; the forward equation, the probability distribution and how it evolves over time. The forward equation is [6]:

$$P'(t) = P(t)G \quad (12)$$

And the backward is [6]:

$$P'(t) = GP(t) \quad (13)$$

The latter being of greater interest for the calculation to be performed. Now, differentiating both sides of the

stationary state equation:

$$\frac{d}{dt} [\pi = \pi P(t)] \quad (14)$$

$$0 = \pi P'(t) \quad (15)$$

$$\pi GP(t) = 0 \quad (16)$$

$$P(t) \neq 0, \therefore \pi G = 0 \quad (17)$$

This means that, unlike discrete systems, where a transition matrix with eigenvalue 1 was observed, which described the stationary state, in continuous systems an eigenvalue of 0 is observed.

3.2.1. Lindblad Master Equation

The Choi-Kraus theorem can be used to obtain a general expression for the time evolution of the density matrix $\rho(t)$. Being a Markov process, $\rho(t + dt)$ depends only on $\rho(t)$.

$$\rho(t + dt) = \sum_{\mu} K_{\mu} \rho(t) K_{\mu}^{\dagger} \quad (18)$$

Since we want this equation to have terms proportional to dt ; this is achieved by establishing a proportionality of the Kraus operators with respect to $\sqrt{dt}$. With this approach, for a quantum system with weak coupling to a Markovian environment, the time evolution of the reduced density operator is defined as follows [1]:

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^{\dagger} - \frac{1}{2} \{L_k^{\dagger} L_k, \rho\} \right) \quad (19)$$

Where H is the Hamiltonian, L_k are jump operators describing the dissipative nature of the system's dynamics and γ_k are the associated dissipation rates. The equation can be rewritten with a Lindblad superoperator, or Liouvillian, as follows:

$$\frac{d\rho}{dt} = \mathcal{L}[\rho] \quad (20)$$

3.2.2. Non-Equilibrium Steady States (NESS)

The effects of dissipation can be better understood by studying the solutions of the Lindblad master equation, which for an initial density matrix $\hat{\rho}(0)$ takes the exponential form [1]:

$$\hat{\rho}(0) = e^{\mathcal{L}t} \hat{\rho}(0) \quad (21)$$

The uniqueness and existence of a non-equilibrium steady state corresponding to the eigenvector of the Lindbladian with zero eigenvalue has been demonstrated:

$$\mathcal{L}[\rho_{ss}] = 0 \quad (22)$$

For systems with dissipative coupling, all eigenvalues of the Lindblad superoperator have non-positive real part, and the state with zero eigenvalue represents the NESS towards which the system asymptotically evolves. Moreover, it is convenient to rewrite the master equation in vector form, such that the superoperator L takes matrix form and the steady state satisfies

$$L|\rho_{ss}\rangle\rangle = 0 \quad (23)$$

Following the structure and behavior of the Lindbladian, it is determined that said superoperator is the equivalent of the generator matrix observed in continuous Markovian systems.

3.3. Metropolis-Hastings

3.3.1. Estimating Observables via Monte Carlo

Now the problem takes a more thermodynamic nature. At a temperature T , the system is in a statistical mixture of eigenstates, which will be established as obeying the canonical ensemble.

$$\rho = \frac{1}{Z} e^{\beta \hat{H}} \quad (24)$$

Where $Z = \sum_{\{s\}} e^{\beta E(\{s\})} = \sum_{\{s\}} \exp\left(\beta J_z \sum_{\langle i,j \rangle} s_i s_j\right)$. In this way, calculating the different thermodynamic quantities can avoid summing over all possible states.

Now, any observable can be calculated with the following relation:

$$\langle \hat{O} \rangle = \text{Tr}(\hat{O} \hat{\rho}) = \frac{1}{Z} \sum_{\{s\}} \langle s_1, \dots, s_N | \hat{O} | s_1, \dots, s_N \rangle e^{-\beta E(\{s\})} \quad (25)$$

Which takes the structure of a Monte Carlo problem. More specifically, this specific approach in which a Monte Carlo method can obtain a sample from any distribution function independent of its complexity or dimension is called a Metropolis-Hastings algorithm, or Markov Chain Monte Carlo (MCMC) [1]. Going into more detail on the associated thermodynamic quantities, the mean energy per spin $\bar{E}$ is [8]:

$$\bar{E} = \langle E \rangle = \frac{1}{2} \langle \sum_{i,j} H_{ij} \rangle \quad (26)$$

Also, the average spin will be calculated, which corresponds to the magnetization:

$$\bar{m} = \langle m \rangle = \frac{1}{N^2} \sum_{(i,j)} s_{ij} \quad (27)$$

Now, even if the heat capacity is defined with a derivative, it is more convenient in this case to define it in a way that can be computed similarly to equation 26.

$$C(T) = \frac{\beta}{T} (\langle E^2 \rangle - \langle E \rangle^2) \quad (28)$$

Furthermore, a "flip" motion rule is defined, which delimits less accessible regions of configuration space. Specifically, this will consist of flipping the spin according to the condition of the Metropolis-Hastings acceptance criterion. For each grid size N , a number of "sweeps" — each corresponding to N attempts to flip a spin — necessary to reach equilibrium will be determined.

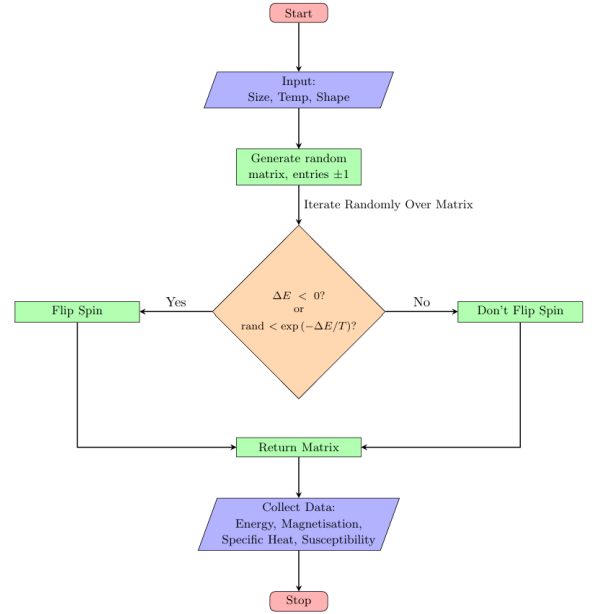


Figure 2: Flow diagram of a sweep for the Metropolis-Hastings algorithm [8]. It should be noted that, in the case of this project, susceptibility was omitted, although it can be calculated analogously to specific heat

3.3.2. Restricted Boltzmann Machines (RBM) and Autocorrelation

It should be mentioned that the main disadvantage of Metropolis-Hastings is due to autocorrelation. That is,

its results depend on the assumption of working with I.I.D's, but in case there is correlation between the distribution variables, this will affect the convergence time of the algorithm. This can alter acceptance rates, and having a very high or low rate negatively affects convergence time. A solution to this problem are Restricted Boltzmann Machines [1].

RBM's are machines that form a family of probability distributions over binary variables, which are separated by two layers: the visible layer and the hidden layer [9]. The visible layer encodes the physical states of the system; the hidden one, the correlations between the degrees of freedom. Also, note that they are defined as restricted because there are no connections within each layer, only between different layers. Then, each wave function ψ_j is represented by an RBM. These machines are based on neural networks, and their training method depends directly on Gibbs sampling [10], which allows obtaining results more efficiently. Simplifying, RBMs allow, through Gibbs sampling and use of neural networks, to train an algorithm on the distribution of interest, and this optimizes the Monte Carlo process with Markov Chains.

However, it is specified that RBMs were not implemented in the project, but their mention is relevant as they provide pertinent theory to the study of open quantum systems, the efficiency of Metropolis-Hastings, and methods when evaluating cases different from the zero external field, thus presenting profound theoretical value.

4. Ising Models

The Ising model was created with the idea of describing ferromagnetism in materials [11]. A system of magnetic spins localized all on a lattice is established such that the spins only interact with the nearest neighbors. It is relevant for studying phenomena such as spontaneous magnetization at low temperatures or magnetic phase transitions. In order to progressively build the necessary theory to study in detail the Ising model in an open quantum system, three Ising models are studied: the classical, the transverse, and finally the transverse dissipative. Due to the extent of theory and methods necessary for the complete study of these systems, a separate section will be dedicated to each.

4.1. Classical Limit of the Transverse Ising Model

Consider an array of spin 1/2 particles with periodic boundary conditions. The Hamiltonian takes the follo-

wing form [1]:

$$\hat{H} = h \sum_i \hat{\sigma}_i^x - J_z \sum_{\langle i,j \rangle} \hat{\sigma}_i^z \hat{\sigma}_j^z \quad (29)$$

Where $\hat{\sigma}_i^\alpha$ are Pauli matrices where $\{\alpha \in x, y, z\}$, h is the uniform transverse field and J_z is the coupling strength. In the case of no coupling with the environment, we have $\gamma = 0$. Studying its relation with the Lindblad equation and the density matrix, its behavior in 1 dimension can be studied.

If we were to work the expression with the external magnetic field, some method such as renormalization of the tensor network ansatz [12], exact diagonalization [13], among others, would be necessary. In order to conduct a more introductory study of this system and reduce the problem to that of the classical 1D Ising model, the external magnetic field will be taken as zero, such that $h = 0$ and the Hamiltonian reduces to the form:

$$\hat{H} = -J_z \sum_{\langle i,j \rangle} \hat{\sigma}_i^z \hat{\sigma}_j^z \quad (30)$$

Now, it is important to consider that the eigenvalues of the Pauli matrices are +1 and -1, and since the summation is performed for each of the particles, the eigenvalue of each Pauli matrix corresponding to each particle will be the spin of that particle. Considering this, the expectation value of the Hamiltonian can be rewritten by performing the following procedure:

$$\begin{aligned} \langle \hat{H} \rangle &= -J_z \sum_{\langle i,j \rangle} \langle \hat{\sigma}_i^z \hat{\sigma}_j^z \rangle \\ \langle \hat{\sigma}_i^z \hat{\sigma}_j^z \rangle &= \langle \psi | \hat{\sigma}_i^z \hat{\sigma}_j^z | \psi \rangle = s_i s_j \langle \psi | \psi \rangle = s_i s_j \\ \therefore \langle \hat{H} \rangle &= -J_z \sum_{\langle i,j \rangle} s_i s_j \end{aligned} \quad (31)$$

With similar logic, besides the expectation value of the Hamiltonian, the total energy can be obtained knowing that it is the eigenvalue of the Hamiltonian:

$$\hat{H}|\psi\rangle = E|\psi\rangle \quad (32)$$

$$\therefore E = -J_z \sum_{\langle i,j \rangle} s_i s_j \quad (33)$$

Therefore, it is determined that in this case $\langle \hat{H} \rangle = E$.

4.2. Transverse Ising Model

The study of quantum phase transitions requires a theoretical framework that transcends classical statistics,

incorporating the non-commutative nature of quantum mechanical operators [14]. The Transverse Field Ising Model (TFIM) is the fundamental archetype for investigating these phenomena, serving as a conceptual and computational bridge between equilibrium statistical mechanics and many-body quantum dynamics. Unlike the classical model discussed in the previous section, where fluctuations are purely thermal, the TFIM introduces quantum fluctuations, driven by an external field transverse to the direction of ferromagnetic ordering.

Using the same definition of a one-dimensional chain with periodic boundary conditions, the Hamiltonian is:

$$\hat{H} = -J_z \sum_{\langle i,j \rangle} \hat{\sigma}_i^z \hat{\sigma}_j^z - h \sum_{i=1}^N \sigma_i^x \quad (34)$$

In this case, the classical limit will not be evaluated; however, it is possible to map the problem of a d -dimensional transverse one to an equivalent classical problem in $d+1$ dimensions. This procedure is based on the equivalence of the time evolution operator $e^{-i\hat{H}t}$ and the unnormalized Boltzmann density operator $e^{-\beta\hat{H}}$ via a Wick rotation. In this rotation, a $t \rightarrow i\tau$ is defined, so the additional dimension corresponds to an imaginary time [14].

The partition function of the quantum system is $Z = \text{Tr}(e^{-\beta\hat{H}})$. Due to the non-commutativity of the terms of the classical Hamiltonian and that associated with the external field, there are problems for performing the standard Metropolis-Hastings process. To solve this, the Suzuki-Trotter expansion [15] is employed, which is detailed in section 4.3.3; M discrete imaginary time intervals are introduced, called "Trotter Slices" of width $\Delta\tau = \frac{\beta}{M}$.

The partition function is rewritten as a discrete path integral over configurations of classical spins $S_{i,k} = \pm 1$, where $i \in [1, N]$ is the spatial index and $k \in [1, M]$ is the temporal index. From this procedure, two effective coupling constants for the resulting anisotropic 2D classical lattice arise. A spatial coupling K is obtained, which is derived from the physical interaction with J , and scales with the time step.

$$K = \Delta\tau J \quad (35)$$

Then, the other constant K_τ corresponds to the temporal coupling, and arises from the matrix elements of the transverse field operator.

$$K_\tau = -\frac{1}{2} \ln(\tanh(\Delta\tau h)) \quad (36)$$

This relation is critical, since as $T \rightarrow 0, \beta \rightarrow \infty$, which causes $K_\tau \rightarrow \infty$. In physical terms, this means that for

low temperatures, the coupling in the imaginary time direction becomes infinitely strong, forcing the spins to form rods along the temporal dimension, interrupted only by quantum jumps, or kinks induced by the field h [14].

4.2.1. Observables and Thermodynamic Estimators

Measuring observables in this formalism requires estimators that average over the additional temporal dimension.

For the longitudinal magnetization m_z , the order parameter is calculated as the average spin over the entire space-time lattice. Since in a finite system the net magnetization can fluctuate between positive and negative due to the inversion symmetry, it is necessary to calculate the root mean square value.

$$|m_z| = \sqrt{\left\langle \left(\frac{1}{NM} \sum_{i,k} S_{i,k} \right)^2 \right\rangle} \quad (37)$$

For the transverse magnetization m_x , the calculation is more difficult since the Pauli matrix is not diagonal. Using the thermodynamic identity $\langle \hat{H}_x \rangle = -h \langle M_x \rangle = \frac{\partial \ln Z}{\partial \beta}$, it can be shown that the transverse magnetization is proportional to the density of "kinks" or defects in the temporal direction. A "kink" occurs when $S_{i,k} \neq S_{i,k+1}$. The implemented estimator is then:

$$\langle m_x \rangle = \frac{\langle N_{kinks} \rangle}{N\beta h} \quad (38)$$

Where N_{kinks} is the total number of times the spins flip along the temporal direction. This inverse relation with h reflects that for very small fields, kinks are infrequent.

4.3. Exact Diagonalization

While the Path Integral Monte Carlo (PIMC) method is a scalable stochastic tool, exact diagonalization provides precise numerical solutions [13]. The procedure consists of constructing the Hamiltonian matrix in a complete basis of the Hilbert space, whose dimension is $D = 2^N$. In this case, $N = 8$ is chosen, which generates a 256×256 matrix.

4.3.1. Construction of Operators via Kronecker Products

To represent many-body operators in matrix form, the Kronecker product is used. A local operator σ_i^α acting on site i is expanded to the complete Hilbert space as:

$$\hat{\sigma}_i^\alpha \rightarrow \mathbb{I}_1 \otimes \mathbb{I}_2 \otimes \cdots \otimes \hat{\sigma}_i^\alpha \otimes \cdots \otimes \mathbb{I}_N \quad (39)$$

At a computational level, this will be performed iteratively.

4.3.2. Sparse Matrices and Krylov Subspaces

A crucial characteristic of the Ising Hamiltonian is that it is a sparse matrix. Most of its elements are zero, since the operator σ_z is diagonal and σ_x only connects states that differ by a single spin. To obtain the ground state $|\psi_0\rangle$ and its energy E_0 , it is not necessary to diagonalize the entire matrix. Iterative algorithms based on Krylov are used. In the computational implementation, quick convergence to the lowest energy state is achieved by projecting the operator onto a reduced-dimensional subspace generated by powers of the Hamiltonian applied to an initial random vector.

4.3.3. Trotter Decomposition

The justification for the imaginary discretization lies in the Lie-Trotter product formula [15]. Given two non-commuting operators A and B , the exponential of their sum is not the product of their exponentials:

$$e^{-\beta(A+B)} \neq e^{-\beta A} e^{-\beta B} \quad (40)$$

However, Trotter's theorem states that equality is asymptotically recovered by dividing the interval β into M infinitesimal segments.

$$e^{-\beta(A+B)} = \lim_{M \rightarrow \infty} \left(e^{-\frac{\beta}{M} A} e^{-\frac{\beta}{M} B} \right) \quad (41)$$

In numerical simulation, M is finite. The first-order approximation, which was employed, introduces a systematic error derived from the commutators of the operators.

$$e^{-\Delta\tau A} e^{-\Delta\tau B} = \exp \left(-\Delta\tau(A+B) + \frac{\Delta\tau^2}{2} + O(\Delta\tau^3) \right) \quad (42)$$

This implies that the error in the effective action per time step is of order $O(\Delta\tau^2)$. Given $M = \frac{\beta}{\Delta\tau}$ steps. The total accumulated error in the partition function scales linearly with step size:

$$\text{Total Error} \propto M \cdot O(\Delta\tau^2) = \frac{\beta}{\Delta\tau} O(\Delta\tau^2) = O(\beta\Delta\tau) \quad (43)$$

This linear dependence is used for convergence analysis. To obtain accurate physical results, it is ideal that M be sufficiently large such that $\Delta\tau \ll 1/J$ and $\Delta\tau \ll 1/h$.

4.3.4. Wolff Algorithm

The standard Metropolis-Hastings algorithm suffers from critical slowing down when approaching the phase transition point. At the point where $h \approx J$, the correlation length diverges such that large domains of aligned spins form. Then, flipping a spin within a large domain entails a high energy cost at the spin boundary, resulting in a low acceptance rate and increasing autocorrelation times as well. To mitigate this, cluster algorithms such as Wolff's are commonly used [16].

For the implementation of the Wolff algorithm, the strong anisotropy between the spatial and temporal directions derived from the Trotter mapping must be considered. The probabilities of adding a neighbor to the cluster depend on the bond directions. There is the spatial bond, which is the probability of adding a spatial neighbor $i \mp 1, k$ with the same sign is $P_{add}^x = 1 - \exp^{-2K}$, where $K = \Delta\tau J$. Then there is the temporal bond, which works analogously; the probability of adding a temporal neighbor $(i, k \mp 1)$ is $P_{add}^\tau = 1 - e^{-2K_\tau}$. For a small $\Delta\tau$, $P_{add}^x \approx 2\Delta\tau J$ and $P_{add}^\tau \approx 1$. This disparity in magnitude causes clusters to grow rapidly along the imaginary time direction, forming elongated structures like needles before expanding spatially. Physically, this means that quantum fluctuations, or cuts in the needle, are uncommon, and spins are strongly correlated in time. This method is notably more efficient regarding autocorrelation reduction.

L	10	12	16	24	32	48	64	96
τ_E , Metropolis	5.79	7.87	12.9	26.5	44.9	93.1	160	336
τ_E , Swendsen-Wang	3.55	3.98	4.69	5.54	6.10	7.50	8.55	10.7
τ_E , Wolff	1.18	1.24	1.31	1.46	1.54	1.71	1.81	1.98

Figure 3: Autocorrelation time of the energy for different τ_E for grids of sizes L

4.4. Dissipative Ising Model

Up to this point, the Ising system has been considered as a closed quantum system. However, in real implementations, it is common that there is a "thermal bath" that introduces decoherence and friction to the system, altering the dynamics of phase transitions. Now, the theory of open quantum systems addresses this problem starting from a Hamiltonian of the form $\hat{H} = \hat{H}_{\text{transversal}} + \hat{H}_{\text{baño}} + \hat{H}_{\text{int}}$ [17]. $\hat{H}_{\text{transversal}}$ is the same Hamiltonian from the previous section, and the other Hamiltonians result from the existence of the bath.

$$\hat{H}_{\text{baño}} = \sum_i \sum_k \omega_{k,i} b_{k,i}^\dagger b_{k,i} \quad (44)$$

Where $b_{k,i}^\dagger$ and $b_{k,i}$ are creation and annihilation operators of mode k coupled to spin i and $\omega_{k,i}$ is the mode frequency. Then, for the interaction Hamiltonian:

$$\hat{H}_{\text{int}} = \sum_i \sigma_i^z \otimes \sum_k g_{k,i} (b_{k,i}^\dagger + b_{k,i}) \quad (45)$$

Where $g_{k,i}$ are coupling constants that determine the spectral density of the bath $J(\omega)$. The form of this term can be ohmic, sub-ohmic, etc., which defines the physics of dissipation. For this problem, the ohmic case is chosen.

4.4.1. Caldeira-Leggett

In this framework, quantum ohmic dissipation is described, which is a friction proportional to velocity in the classical limit [18]. Applying this formalism to the Ising model, the Euclidean action is augmented by a long-range interaction term in imaginary time.

$$S_{\text{eff}} = S_{\text{Ising}} + \frac{\alpha}{2} \sum_i \int_0^\beta d\tau \int_0^\beta d\tau' K(\tau - \tau') (S_i(\tau) - S_i(\tau'))^2 \quad (46)$$

The parameter α quantifies the dissipation strength. The interaction kernel $K(\tau - \tau')$ decays asymptotically as the inverse square of the temporal distance. Physically, this term penalizes configurations where $S_i(\tau) \neq S_i(\tau')$. That is, a long-range ferromagnetic stiffness in time is introduced that suppresses kinks, or quantum tunneling. While the transverse field attempts to create fluctuations, or kinks, the dissipation α seeks to join and annihilate them, localizing the spin in a fixed state.

Given that imaginary time is periodic, the standard distance $|\tau - \tau'|$, which is the periodic kernel, must be replaced by a metric that respects the circular topology of the interval $[0, \beta]$. The metric is the chord distance, which corresponds to the Euclidean distance between two points on a circle of circumference β .

$$d(t, t')^2 = \left(\frac{M}{\pi} \sin \left(\frac{\pi |t - t'|}{M} \right) \right)^2 \quad (47)$$

For $|t - t'| \ll M$, it approximates $|t - t'|^2$, which recovers the Caldeira-Leggett limit, but guarantees exact periodicity without discontinuities at the boundary $\tau = \beta$.

Now, similar to the transverse Ising case, global updates are necessary to prevent freezing. However, in this

case, the Wolff method does not apply, and furthermore, the source of freezing is more related to the dissipative term, which creates an energy landscape with high barriers for local changes. A completely local algorithm would be extremely inefficient for large α , so the process would get trapped in local minima. Therefore, to counteract this phenomenon, the temporal rod of a spin is flipped simultaneously $S_{i,t} \rightarrow -S_{i,t} \forall t$. The internal energy of the rod is invariant under a global inversion, and only changes through the spatial interaction energy with the neighbors. In this way, different sectors of global magnetization can be explored even when dissipation is so strong that it suppresses almost all individual quantum fluctuations, allowing correct simulation of the localization transition induced by the environment predicted by Schmid [19].

5. Methodology

Primarily, it is worth highlighting how the Metropolis-Hastings algorithm takes a leading role, such that the study of the Ising model dynamics cannot be performed without the numerical solutions obtained from said method. Specifically, it refers to equation 25, with which the expectation value of an arbitrary observable is defined by approximating the problem to a Monte Carlo one.

5.1. Classical Ising

5.1.1. Generation of the one-dimensional grid

An array of N spins with values $s_i \in \{-1, +1\}$ is generated via uniform probability. The system is configured with periodic boundary conditions: $s_{N+1} = s_1$.

5.1.2. Initial energy calculation

The initial energy of the system is calculated according to:

$$E_0 = -J_z \sum_{\langle i,j \rangle} s_i s_j \quad (48)$$

5.1.3. System parameters

- Coupling constant: $J_z = 1 \times 10^{-21}$ J, which describes a ferromagnetic system. This value was set

two orders of magnitude above Boltzmann's constant, to avoid overflow error and also to have sufficiently substantial results that are visible, to facilitate analysis.

- Thermal bath temperature: T (K),
- Boltzmann factor: $\beta = \frac{1}{k_B T}$

5.2. Metropolis-Hastings Algorithm

For each iteration $t = 1, 2, \dots, M$, the following steps are executed:

5.2.1. Movement proposal

1. Randomly select a site $i \in [1, N]$
2. Propose spin flip: $s_i \rightarrow -s_i$

5.2.2. Local energy change calculation

The energy change associated with the proposed flip is:

$$\Delta E = 2J_z s_i (s_{i-1} + s_{i+1}) \quad (49)$$

where periodic boundary conditions are applied at the system boundaries.

5.2.3. Acceptance criterion

The probability of accepting the proposed movement is given by the Metropolis criterion:

$$P_{\text{accept}} = \begin{cases} 1 & \text{if } \Delta E \leq 0 \\ e^{-\beta \Delta E} & \text{if } \Delta E > 0 \end{cases} \quad (50)$$

A random number $r \sim U(0, 1)$ is generated:

- If $r < P_{\text{accept}}$: the move is accepted, updating s_i and $E \rightarrow E + \Delta E$
- If $r \geq P_{\text{accept}}$: the move is rejected, maintaining the current configuration

5.2.4. Observable recording

At each step t , the total energy $E(T)$ is recorded:

5.3. Thermalization

The first N_{term} iterations are discarded to allow the system to reach thermal equilibrium. Thermodynamic observables are calculated using only the last $N_{\text{eq}} = 10000$ equilibrium configurations.

5.4. Calculation of Thermodynamic Quantities

For each temperature T_k in the range $[T_{\text{mín}}, T_{\text{máx}}]$ with step $\Delta T = 5$ K, the following quantities are calculated:

5.4.1. Average energy per spin

$$\langle E \rangle = \frac{1}{N \cdot N_{\text{eq}}} \sum_{t=M-N_{\text{eq}}}^M E(t) \quad (51)$$

5.4.2. Average magnetization per spin

$$\langle m \rangle = \frac{1}{N \cdot N_{\text{eq}}} \sum_{t=M-N_{\text{eq}}}^M M(t) \quad (52)$$

Which is an equivalent way of writing equation 26, but in a way that can be computed with the described methodology.

5.4.3. Specific heat

$$C(T) = \frac{\text{Var}[E]}{N k_B T^2} \quad (53)$$

where $\text{Var}(E) = \langle E^2 \rangle - \langle E \rangle^2$.

5.5. Transverse Ising Model (PIMC)

For the study of the Transverse Field Ising Model, the classical Monte Carlo methodology was extended through the path integral formalism. Computationally, this implied the simulation of a two-dimensional lattice of dimensions $N \times M$, where N represents spatial positions and M the imaginary time intervals (Trotter slices).

5.5.1. Initialization and Effective System Parameters

The simulation is initialized by generating a random configuration of spins $s_{i,k} \in \{-1, +1\}$ in a matrix of size (N, M) . Unlike the isotropic classical case, the effective 2D lattice presents anisotropic couplings derived from the Suzuki-Trotter expansion.

The following computational parameters were defined based on the imaginary time discretization $\Delta\tau = \beta/M$:

- **Spatial Coupling (K):** Determined by the ferromagnetic interaction scaled by the time step, $K = \Delta\tau J$.
- **Temporal Coupling (K_τ):** Determined by the transverse field intensity, calculated as:

$$K_\tau = -\frac{1}{2} \ln(\tanh(\Delta\tau h)) \quad (54)$$

It is important to note that for small transverse fields ($h \rightarrow 0$), the temporal coupling K_τ tends to infinity, which forces the formation of continuous ferromagnetic rods in the temporal direction, meaning that quantum fluctuations disappear. In the case of temperatures, only the low-temperature case is chosen to be studied computationally, since the high-temperature case presents a very high error from the Trotter method. In the low-temperature limit, $K_\tau \rightarrow +\infty$, so the same as said for a small transverse field is repeated. For high temperatures $K_\tau \rightarrow 0$, so the temporal dimension collapses, preventing the formation of order and creating a state of maximum entropy.

5.5.2. Update Algorithms: Local and Wolff

To guarantee ergodicity and efficient convergence, especially near the quantum critical point ($h \approx J$), a hybrid update strategy that interleaves local sweeps with cluster updates was implemented.

1. Local Update: A spin $s_{i,t}$ is randomly selected. The energy change ΔE considers both spatial ($i \pm 1, t$) and temporal ($i, t \pm 1$) neighbors. Acceptance is governed by the standard Metropolis criterion:

$$P_{\text{accept}} = \min(1, e^{-\Delta E_{\text{total}}}) \quad (55)$$

2. Wolff Update (Clusters): Since local updates suffer from high autocorrelation in the ordered phase, the Wolff algorithm adapted to the anisotropic lattice was programmed. Cluster growth is performed using a stack, adding neighbors with probabilities dependent on the bond direction:

- **Spatial addition probability:** $P_{\text{add}}^x = 1 - e^{-2K}$
- **Temporal addition probability:** $P_{\text{add}}^\tau = 1 - e^{-2K_\tau}$

The code executes one Wolff update for every 10 complete local update sweeps.

5.5.3. Calculation of Quantum Observables

Measurement of physical quantities was performed after a thermalization period of 20 % of the total steps (N_{sweeps}).

- **Longitudinal Magnetization:** Calculated as the square root of the average of the square of the instantaneous magnetization to avoid cancellation by symmetry:

$$|m_z| = \sqrt{\langle (\frac{1}{NM} \sum_{i,k} s_{i,k})^2 \rangle} \quad (56)$$

- **Transverse Magnetization (m_x):** Since the operator σ^x is not diagonal in the z basis, its expectation value was estimated via the density of "kinks" or jumps in the temporal direction. Computationally, pairs where $s_{i,t} \neq s_{i,t+1}$ are identified and the estimator is applied:

$$\langle m_x \rangle = \frac{\langle N_{\text{kinks}} \rangle}{N\beta h} \quad (57)$$

To validate the implementation, an analytical benchmark module was developed using exact diagonalization for small systems ($N = 8$), comparing the energy density and magnetization obtained by PIMC with the eigenvalues of the Hamiltonian constructed via Kronecker products of Pauli matrices.

5.6. Dissipative Ising Model (Caldeira-Leggett)

For the study of decoherence induced by the environment, the PIMC algorithm was modified to include the ohmic dissipation term. This introduces additional computational complexity, since the interaction in imaginary time ceases to be nearest-neighbor and becomes a long-range interaction.

5.6.1. Memory Kernel Calculation

Since the dissipative interaction connects all time slices to each other, it was chosen to pre-compute the kernel

matrix $K(\tau - \tau')$ at the start of the simulation to optimize performance.

To respect periodic boundary conditions in β , the "chord distance" metric was implemented in the calculation of the interaction. The matrix element for two instants t and t' is defined in the code as:

$$K_{t,t'} = \frac{\alpha \pi^2}{M^2 \sin^2\left(\frac{\pi|t-t'|}{M}\right)} \quad (58)$$

Where α is the dimensionless parameter controlling the dissipation intensity.

5.6.2. Global Updates for Freezing Prevention

The introduction of α generates high energy barriers that suppress quantum fluctuations (kinks), which can trap the local Metropolis algorithm in metastable local minima. To mitigate this ergodicity problem, a type of global movement called "Uniform Rod Update" was implemented.

This movement proposes to invert the spin of the entire temporal column corresponding to a spatial site i simultaneously ($s_{i,t} \rightarrow -s_{i,t}, \forall t \in [1, M]$).

- The change in internal energy of the rod due to dissipation and the transverse field is zero (inversion symmetry).
- The only energy change comes from the spatial interaction with neighbors $i - 1$ and $i + 1$.

This technique allows the system to explore different global magnetization sectors even in high friction regimes ($\alpha > 1$), where individual quantum tunneling is strongly suppressed.

5.6.3. Phase Transition Measurement Protocol

To characterize the localization transition, a sweep of the parameter α was performed in the range $[0, 0.2, 0]$ keeping fixed $N = 8$, $\beta = 10.0$ and a transverse field $h = 1.5J$ (corresponding to the disordered phase in the closed case).

A temporal discretization of $M = 60$ slices was used and 3000 sweeps per point were executed, discarding the initial 30 % for thermalization. The order parameter $|m_z|$ and quantum fluctuations m_x were simultaneously recorded to visualize the crossover between the quantum fluctuation regime and the localized phase induced by the quantum Zeno effect of the environment.

5.7. Results Analysis

The following quantities are graphed:

1. **Temporal evolution:** $E(t)$ and $M(t)$ versus iterations, to verify system thermalization. The high temperature range ($T = 10000K$) and low temperature range ($T = 1K$) will be studied.
2. **Thermal dependence:** $\langle E \rangle(T)$, $\langle m \rangle(T)$ and $C(T)$ in the studied temperature range

Convergence to equilibrium is verified by:

- Stabilization of $E(t)$ and $M(t)$ after the thermalization period
- Statistically stable fluctuations in the equilibrium regime

In addition to the computational procedure, the obtained results will be used to study the behavior of equation 19 at high and low temperatures, so that both results, theoretical and computational, complement each other. For this, a direct connection will be made between the energy in the steady state for a specific temperature and the coherence and decoherence of the temporal evolution of the ansatz.

6. Results: Classical Ising Model

The complete study is performed for 25 spins, which is sufficient to observe the average behavior given the initialization of the spins according to the uniform distribution.

6.1. Total energy, average spin and specific heat

6.1.1. Low temperature limit

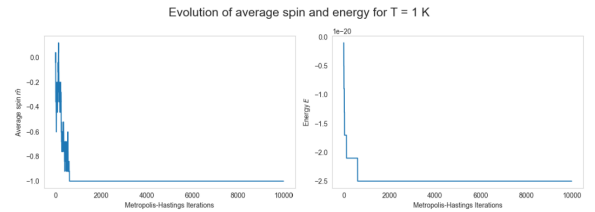


Figure 5: Evolution of average spin and total energy for a temperature of $1K$, calculated via Metropolis Hastings

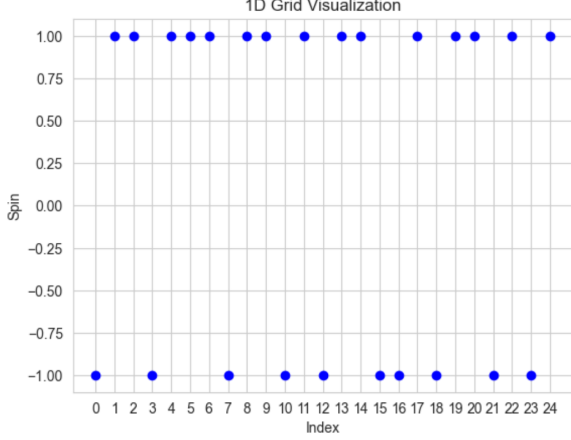


Figura 4: Example of the visualization of 25 spins (+1, -1) distributed on the grid

Note that the average spin seen in figure 5 eventually converged to the value of -1 , meaning that all spins in the system aligned in the same direction. This has a fairly intuitive physical meaning, which is that of energy minimization. When energy is minimized, the system will be in the ground state, so it will seek to minimize the Hamiltonian, which only happens when $s_i, s_j = 1$ or $s_i, s_j = -1$, so when multiplying them that average value will always result in ∓ 1 .

Then, for the total energy, having aligned spins it is easy to directly calculate the summation according to equation 26, such that it can be determined that $\bar{E} = -J \times 25 = -2,5 \times 10^{-20}$, which corresponds to what was calculated.

6.1.2. High temperature limit

For high energies, a highly entropic system will be present, causing constant flipping of the spins. Therefore, it can be expected that for a sufficiently large number of spins, they will be flipped uniformly at any moment, so that the average spin is 0. However, since the system remains highly entropic, the average spin will always oscillate around said value, which is observed in image 6. This behavior transfers to the total energy, which in turn depends on the spins, and will also oscillate around 0.

In order to perform a more in-depth theoretical analysis, and relate it more directly to the quantum definition of the Ising model, this can be explained by returning to the definition of the Hamiltonian without an external magnetic field and the calculation of an observable via a trace. It had been previously determined that $E = \langle \hat{H} \rangle$. Now, for the high temperature limit, there is the same

probability of finding the system in any state, which will be represented in the ansatz. Thus, the total energy can be written as follows:

$$E = \langle \hat{H} \rangle = \text{Tr}(\rho \hat{H}) = \frac{1}{2^N} \text{Tr}(\hat{H})$$

From equation 30, the Pauli matrices for each spin are related via a tensor product, so the trace identity and the fact that the trace of the Pauli matrices is 0 can be used to perform the following:

$$\begin{aligned} \text{Tr}(\hat{H}) &= -J_z \sum_{\langle i,j \rangle} \text{Tr}(\hat{\sigma}_i^z \hat{\sigma}_j^z) \\ \text{Tr}(\hat{\sigma}_i^z \hat{\sigma}_j^z) &= \text{Tr}(\hat{\sigma}_i^z) \text{Tr}(\hat{\sigma}_j^z) = 0 \\ \therefore E &= 0 \end{aligned}$$

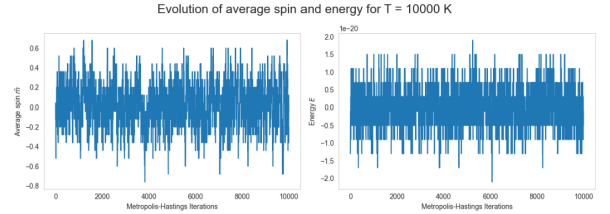


Figure 6: Evolution of average spin and total energy for a temperature of 10000K, calculated via Metropolis Hastings

6.1.3. Total energy and specific heat according to temperature

Note how, in the case of total energy, the change from one limit to the other can be seen, as observed in figures 6 and 5, which additionally corroborates the results obtained in those sections.

It can be observed how specific heat increases as energy increases. This can be explained because, as this increase occurs, the energy states are "filled", making it more difficult for the system to distribute the additional energy that enters. Therefore, a saturation effect occurs, such that increasingly more energy is required to raise the temperature by one degree.

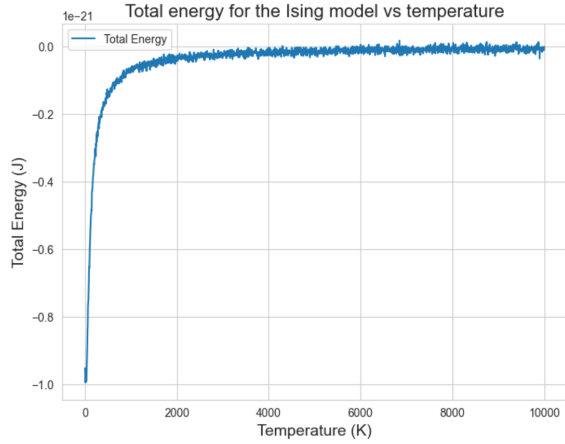


Figura 7: Total energy of the 1D Ising model according to temperature

7. Results: Transverse Ising Model

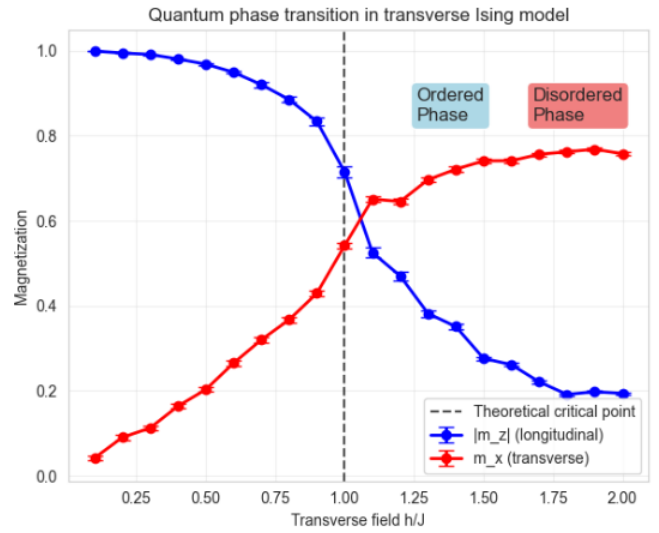


Figure 9: Magnetization vs. transverse field for the transverse Ising model, showing the phase transition

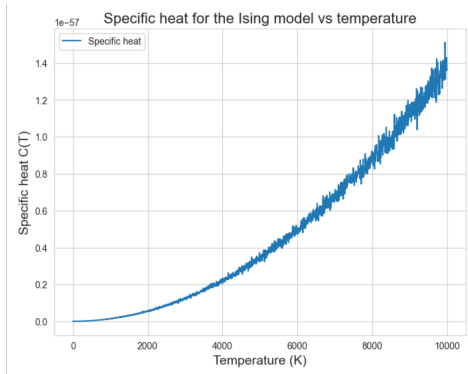


Figure 8: Specific heat of the 1D Ising model according to temperature

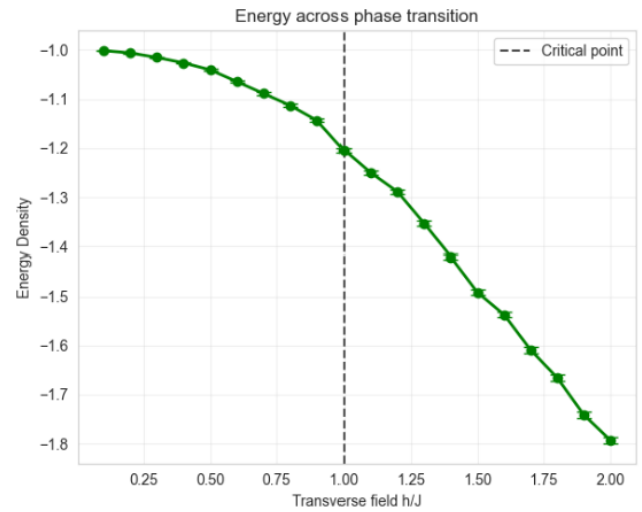


Figure 10: Relation of longitudinal and transverse magnetization intensity vs. dissipation intensity for the Ising model in an open quantum system

For the study of the Transverse Ising Model, a sweep of the transverse magnetic field was performed in a range from $0,1J$ to $2,0J$, maintaining an effective temperature close to zero to ensure that the observed fluctuations are purely quantum and not thermal in nature.

When analyzing the behavior of longitudinal and transverse magnetization as a function of field intensity, a clear second-order phase transition is observed at the critical point $h_c \approx 1J$.

In the ordered ferromagnetic region, where the transverse field is weak, the interaction term dominates the Hamiltonian dynamics. Computationally, this is reflected in the temporal coupling K_τ . Recalling the discussion in the theoretical framework, it is important to mention how the case of zero external field was evaluated since it is known from the established theory that, in such a case, the temporal coupling is infinite.

Physically, this implies that spins are strongly correlated along imaginary time. The energy cost of creating a "kink" is prohibitive. Consequently, temporal rods remain rigid and spatially aligned, breaking inversion symmetry and resulting in a parameter such that the absolute longitudinal magnetization tends to 1. Given that a low temperature was chosen, this corroborates the result obtained for classical Ising, thus successfully replicating the known results in the transverse model.

For the disordered paramagnetic phase, upon surpassing the critical point, an abrupt collapse of longitudinal magnetization is observed. In this regime, the external field term dominates the total energy. From the perspective of quantum mechanics, the transverse field induces intense quantum fluctuations that cause tunneling between states.

It is notable to observe how the behavior of transverse magnetization is proportional to the density of kinks in the temporal dimension. The increase of this quantity indicates that temporal rods are being "cut" frequently by quantum fluctuations, destroying long-range order. The ground state becomes a unique state in the σ_x basis, where the expectation value of σ_z is zero.

8. Results: Transverse Dissipative Ising Model

Once the quantum phase transition was characterized in the closed system, the effect of an environment of ohmic nature was studied. The system was fixed in a disordered regime, where $h = 1.5J$ and the dissipation intensity α was varied. The recovery of magnetic order through system friction was observed, which is a counterintuitive behavior from a classical perspective, but relevant in the study of open quantum systems.

In figure 11, it is observed that for low values of α , the system remains in the paramagnetic phase dominated by the transverse field. However, upon crossing a critical threshold, which was found around $\alpha_c \approx 0.05$ for the parameters used, the longitudinal magnetization grows rapidly until saturating at unity, while transverse fluctua-

tions are suppressed to zero. This is explained by analyzing the effective action of equation 46. The dissipative term introduces a long-range ferromagnetic interaction in imaginary time, which energetically penalizes any change in spin configuration along the temporal trajectory. It is worth mentioning how at the point without dissipation, the obtained result corresponded to what was observed in the transverse Ising model, corroborating the veracity of the results. Thus, when there is no dissipation, the Hamiltonian allows coherences and superpositions, facilitating quantum tunneling driven by σ_x . On the other hand, when there is high dissipation, the interaction with the environment destroys the phase coherence necessary for tunneling of the potential wells. The system suffers localization, becoming trapped in one of the ground states of the interaction Hamiltonian.

Therefore, the localization transition predicted by the theory of Schmid and Caldeira-Leggett was numerically confirmed. Despite the presence of an external field that seeks to disorder the system, friction restores classical ferromagnetic order at low temperatures.

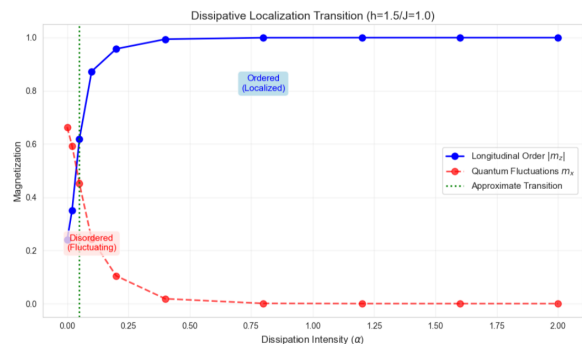


Figure 11: Relation of longitudinal and transverse magnetization intensity vs. dissipation intensity for the Ising model in an open quantum system

9. Temporal evolution of the ansatz for high and low temperature limits

Now, we can connect the steady states seen in the previous section with the temporal dynamics that produce them, which is described by the Lindblad master equation. This evolution consists of a coherent term, coming from the Hamiltonian ($-i[H, \rho]$) and a dissipative one, describing the decoherence and relaxation imposed by the environment ($\sum_k \dots$)

9.0.1. Low Temperature Limit ($T \rightarrow 0$)

In this regime, the Metropolis-Hastings results seen in figure 5 show that the system converges to a minimum energy state, with all spins aligned, resulting in high magnetization. Analyzing the temporal evolution, a system starting from an arbitrary initial state, such as a quantum superposition, will experience two effects from the Lindblad dynamics.

First, the dissipative term $\sum_k \dots$ will induce rapid decoherence. The thermal bath "measures" the system, causing a collapse of quantum superpositions (the coherences) into the energy basis. Since the Hamiltonian described in equation 30 is diagonal in the σ^z basis, this is the preferred basis. Once the state is a statistical mixture, or a diagonal state without coherences, the same dissipative term continues to act, but now as a relaxation mechanism (a "cooler"), extracting energy from the system until it reaches the minimum energy state.

The steady state or NESS, ρ_{ss} , is then the ground state, or a statistical mixture of the degenerate ground states, $|++\dots\rangle$ and $|--\dots\rangle$. In this final state, the coherent term $-i[H, \rho_{ss}]$ vanishes, since the steady state commutes with the Hamiltonian. Therefore, the evolution is completely dominated by dissipation, which guides the system towards this pure and ordered state.

9.0.2. High Temperature Limit ($T \rightarrow \infty$)

For the case of $T = 10000$ K, the computational results seen in figure 6 show a high entropy state, where energy and average magnetization fluctuate around zero.

From the perspective of temporal evolution, in this limit the thermal bath acts as an overwhelming noise source. The dissipative term $\sum_k \dots$ in equation 19 becomes extremely strong and chaotic, representing thermal fluctuations that randomize the system. In this scenario, decoherence is practically instantaneous; any initial quantum coherence is immediately destroyed by thermal noise. Unlike the cold case, the bath not only destroys coherence, but actively injects noise, preventing any form of order.

The Hamiltonian term $-i[H, \rho]$ becomes completely irrelevant, since the thermal energy $k_B T$ is orders of magnitude greater than the coupling energy J_z . The temporal evolution, therefore, leads very quickly to the NESS, which is the maximally mixed state: $\rho_{ss} = \mathbb{1}/2^N$. This maximum entropy state is the only stable solution for $\mathcal{L}[\rho_{ss}] = 0$ under the influence of infinite noise. This NESS predicts exactly $\text{Tr}(\rho_{ss} \hat{H}) = 0$ and $\text{Tr}(\rho_{ss} \hat{M}) = 0$, theoretically validating the computationally obtained results.

10. Temporal evolution of the ansatz for high and low temperature limits

Now, we can connect the steady states seen in the previous section with the temporal dynamics that produce them, which is described by the Lindblad master equation. This evolution consists of a coherent term, coming from the Hamiltonian ($-i[H, \rho]$) and a dissipative one, describing the decoherence and relaxation imposed by the environment ($\sum_k \dots$).

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The steady state or NESS, ρ_{ss} , is then the ground state, or a statistical mixture of the degenerate ground states, $|++\dots\rangle$ and $|--\dots\rangle$. In this final state, the coherent term $-i[H, \rho_{ss}]$ vanishes, since the steady state commutes with the Hamiltonian. Therefore, the evolution is completely dominated by dissipation, which guides the system towards this pure and ordered state.

This cooling and localization dynamics theoretically observed in Lindblad correlates directly with the results obtained in the Dissipative Ising Model (Section VIII). In the PIMC simulation, the Caldeira-Leggett term introduces a long-range interaction in imaginary time that penalizes "kinks" or quantum fluctuations. By increasing dissipation α (Figure 11), we observe how transverse magnetization m_x is suppressed (analogous to the destruction of off-diagonal coherences) and longitudinal magnetization $|m_z|$ is recovered (analogous to relaxation towards the diagonal ground state). This confirms that, in the low temperature limit, the environment acts by imposing an effect that "freezes" the system's dynamics in its preferred subspace and suppresses quantum tunneling induced by the transverse field h .

10.0.2. High Temperature Limit ($T \rightarrow \infty$)

For the case of $T = 10000$ K, the computational results seen in figure 6 show a high entropy state, where energy and average magnetization fluctuate around zero.

From the perspective of temporal evolution, in this limit the thermal bath acts as an overwhelming noise source. The dissipative term $\sum_k \dots$ in equation 19 becomes extremely strong and chaotic, representing thermal fluctuations that randomize the system. In this scenario, decoherence is practically instantaneous; any initial quantum coherence is immediately destroyed by thermal noise. Unlike the cold case, the bath not only destroys coherence, but actively injects noise, preventing any form of order.

The Hamiltonian term $-i[H, \rho]$ becomes completely irrelevant, since the thermal energy $k_B T$ is orders of magnitude greater than the coupling energy J_z . The temporal evolution, therefore, leads very quickly to the NESS, which is the maximally mixed state: $\rho_{ss} = \mathbb{1}/2^N$. This maximum entropy state is the only stable solution for $\mathcal{L}[\rho_{ss}] = 0$ under the influence of infinite noise. This NESS predicts exactly $\text{Tr}(\rho_{ss} \hat{H}) = 0$ and $\text{Tr}(\rho_{ss} \hat{M}) = 0$, theoretically validating the computationally obtained results.

This description is consistent with what was observed in the Suzuki-Trotter expansion used in sections 4 and 8. In the high temperature limit ($\beta \rightarrow 0$), the extension of the imaginary temporal dimension collapses. This implies that temporal correlations induced both by the transverse field and by the Caldeira-Leggett memory kernel become irrelevant compared to the thermal scale. In numerical simulation, this translates into the Metropolis acceptance criterion tending to 1, allowing the system to randomly explore the entire phase space without being restricted by the energy barriers of the Hamiltonian nor by the localization induced by dissipation. Thus, the open quantum system at high temperatures recovers the behavior of a classical system dominated by maximum thermal disorder.

Certainly, here is the conclusions section written in prose, maintaining the academic and formal tone you have used throughout the report, connecting the simulation achievements with the discussed physical theory.

11. Conclusions

In this project, Monte Carlo methods were successfully implemented, specifically the Metropolis-Hastings algorithm and its Path Integral extension (PIMC), for the study of open quantum systems under the dynamics of the Lindblad master equation. For this, a classical and

a transverse Ising model were previously studied, whose results were relevant for corroboration of results and verification with the associated theory. The thermodynamic behavior of the one-dimensional Ising model in the asymptotic temperature limits was verified, observing clear convergence towards the ordered ground state in the low-temperature regime and towards a maximum entropy state in the high-temperature regime, which validates the consistency between numerical simulation and theoretical predictions about the steady-state density matrix (NESS).

Additionally, the quantum phase transition induced by a transverse field was characterized and, crucially, the effect of ohmic dissipation on the system dynamics was demonstrated. The obtained results confirm that interaction with the environment, modeled through the Caldeira-Leggett formalism, induces a localization phenomenon that suppresses quantum fluctuations and restores ferromagnetic order even in the presence of a disordering transverse field. In this way, the stated objectives were fully met by linking computational statistical mechanics with the phenomenology of decoherence and the quantum Zeno effect, providing a robust description of the competition between coherence, thermal fluctuations, and dissipation.

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